Sachin Ashok

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LinkedIn Google Scholar GitHub

Research Statement

I am a researcher specializing in networked systems, focusing on improving the **performance, reliability, and observability of cloud and microservice infrastructure**. My work combines systems design, large-scale experimentation, and algorithmic techniques to solve challenges in distributed request tracing and request scheduling within complex distributed environments, developing deployable solutions that require no code modifications to existing services. The resulting work has led to publications at top tier networking venues such as SIGCOMM, SIGMETRICS, and HotNets.

Education

2021 – 2026 **Ph.D. in Computer Science, GPA: 4.0** Expected 12/26
University of Illinois Urbana-Champaign
Advisors: **Prof. Radhika Mittal**, **Prof. Philip Brighten Godfrey**

2014 – 2018 **B.Tech. in Computer Science and Engineering**
National Institute of Technology, Trichy

Industry Experience

2024 **Microsoft Research, Redmond** **Research Intern**
Mentor: **Prateesh Goyal**.
Explored adaptive scheduling and admission control techniques to improve efficiency of a telemetry pipeline.

2018 – 2021 **Microsoft Research, India** **Research Fellow**
Mentors: **Venkat Padmanabhan**, **Nagarajan Natarajan**, **Johannes Gehrke**.
Designed a data-informed network simulator for recreating the behavior of paths in a target network.

2017 **Samsung R&D, India** **Intern**
Mentors: **Pratibha Moogi**, **Karthikeyan Somanathan**.
Developed an end-to-end object localization framework using deep learning models like CNNs.

2016 **Mozilla Organization** **Intern**
Mentors: **Franziskus Kiefer**, **Tim Taubert**.
Implemented Blake2 (a cryptographic hash function) support for Mozilla's NSS library.

Publications

[1] **TraceWeaver: Distributed Request Tracing for Microservices Without Application Modification**
Sachin Ashok, Vipul Harsh, Brighten Godfrey, Radhika Mittal, Srinivasan Parthasarthy, Larisa Shwartz
In Proceedings of ACM SIGCOMM, 2024.

[2] **Murphy: Performance Diagnosis of Distributed Cloud Applications**
Vipul Harsh, Wenxuan Zhou, Sachin Ashok, Radhika N. Mysore, Brighten Godfrey, Sujata Banerjee
In Proceedings of ACM SIGCOMM, 2023.

[3] **Data-Driven Network Path Simulation with iBox**
Sachin Ashok, Shubham Tiwari, Nagarajan Natarajan, Venkat Padmanabhan, Sundararajan Sellamanickam
In Proceedings of ACM SIGMETRICS, 2022.

[4] **Fast and Efficient Look-Ups via Data-Driven FIB Designs**
Sachin Ashok*, Aditi Partap*, Ammar Tahir* (* = equal contribution)
In Proceedings of ACM SIGCOMM FIRA Workshop, 2022.

[5] **Leveraging Service Meshes as a New Network Layer**
Sachin Ashok, P. Brighten Godfrey, Radhika Mittal
In Proceedings of ACM HotNets, 2021.

[6] **iBox: Internet in a Box**
Sachin Ashok, Sai Surya Duvvuri, Nagarajan Natarajan, Venkat Padmanabhan, Sundararajan Sellamanickam, Johannes Gehrke
In Proceedings of ACM HotNets, 2020.

[7] **RL for Bandwidth Estimation and Congestion Control in Real-Time Communications**
Joyce Fang, Martin Ellis, Bin Li, Siyao Liu, Yasaman Hosseinkashi, Michael Revow, Albert Sadovnikov, Ziyuan Liu, Peng Cheng, **Sachin Ashok**, David Zhao, Ross Cutler, Yan Lu, Johannes Gehrke
In Proceedings of NeurIPS MLForSystems Workshop, 2019.

In Preparation

[1] **Surfacing Latent Application Insights for an Optimized Cloud**
Sachin Ashok*, Karuna Grewal*, Gangmuk Lim, Talha Waheed, Gerard Matthew, Aditya Prerepa, Brighten Godfrey, Justin Hsu, Radhika Mittal
In Preparation.

[2] **Enabling Slack Based Scheduling for Microservice Systems**
Sachin Ashok, Brighten Godfrey, Radhika Mittal
In Preparation.

[3] **TierX: Adaptive Prioritization to Meet SLO Guarantees for Diverse Workloads**
Gerard Matthew, **Sachin Ashok**, Brighten Godfrey, Radhika Mittal
Under Submission.

[4] **Multiparty Load Balancing**
Talha Waheed, **Sachin Ashok**, Brighten Godfrey, Radhika Mittal, R. Srikanth
In Preparation.

Invited Talks

Distributed Tracing without the Pain!
KubeCon 2022, Detroit, USA

iBox: Internet in a Box for Realistic Network Simulation
Microsoft Bandwidth Control Workshop 2020, Microsoft Research, Redmond

Academic Service

External Reviewer, SIGCOMM 2025, IEEE Network Magazine 2024
Artifact Evaluation Committee, SIGCOMM 2025
Teaching Assistant, CS435 - Cloud Networking (Spring 2025)
Guest Lecture, CS425 - Distributed Systems (Spring 2024)
Guest Lecture, ECE598MCI - Modern Cloud Infrastructure (Spring 2025)
Mentor, Research Fellow Mentorship Committee 2024, Microsoft Research India
Invited Panelist, NetworkingChannel 2024

Awards and Honors

Student Grants and Travel Awards, SIGMETRICS 2022, SIGCOMM 2023, HotNets 2023
Project Demo Invite, Microsoft TechFest, Seattle, 2020
Finalist, Pragyan CTF, NIT Trichy (Top 1% of participants), 2018
Selected Participant, IISc CSA Summer School (Top 100 chosen nationwide), 2017
Selected Participant, IMSc Summer Research Program (Top 25 chosen nationwide), 2017
Selected Participant, Mozilla Winter of Security (Among 12 teams chosen globally), 2016

Skills and Background

Languages: C++, C, Python, Bash (*advanced*); Golang, JavaScript (*proficient*)
Tools: Kubernetes, Docker, eBPF, Istio, Protobuf, ns-2/3, gRPC, TensorFlow, NetEm, Wireshark
Courses Taken: Advanced Networking (CS 538), Advanced Operating Systems (CS 523), Reliability of Cloud Systems (CS 598XU), High-Speed and Programmable Networks (CS 598RM), Advanced Distributed Systems (CS 525)